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Towards an evolutionary approach to learning from assumptions: Lessons from the evaluation of Dancing with Parkinson's

April Nakaima [a,*], Sanjeev Sridharan [b], Rachael Gibson [a]

[a] The Evaluation Centre for Complex Health Interventions, Dalla Lana School of Public Health, University of Toronto, 155 College St Room 500, Toronto ON M5T 3M7, Canada

[b] Social Science Research Institute, University of Hawaii at Mānoa, United States

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ABSTRACT

This paper highlights how learnings from exploring assumptions can be strengthened by taking an *evolutionary approach* to theory building and analysis. We discuss theory-driven evaluation applied to a community-based intervention implemented by Dancing With Parkinson's in Toronto, Canada, targeting Parkinson's disease (PD), a neurodegenerative condition affecting movement. A major gap in the literature is understanding the mechanisms by which dance might make a difference in the daily lives of people living with PD. This study was an early exploratory evaluation to better understand mechanisms and short-term outcomes. Conventional thinking generally favors "permanent" over "transitory" changes, and "long-term" over "short-term" effects. Yet, for people living with degenerative conditions (and also people experiencing chronic pain and other chronic symptoms), transitory and short-term changes may be highly valued and welcomed relief. In order to study and link multiple longitudinal events to explore key linkages in the theory of change, we piloted the use of diaries, with brief entries filled out daily by participants. The aim was to better understand the short-term experiences of participants using their daily routines as a means of learning about potential mechanisms, what matters to participants, and to see if small effects could be observed on days when participants danced versus days when they did not dance and also longitudinally over several months. Our initial theoretical stance began with a view of dance as exercise and the well-established benefits of exercise; yet, we explored through the diary data collected, as well as client interviews and literature review, potential other mechanisms of dancing (such as group interaction, touch, stimulation by the music, and esthetics including "feeling lovely"). This paper does not develop a full, comprehensive theory of dance but moves towards a more comprehensive view that locates dance within the routine activities of participants' daily lives. We argue that given the challenges of evaluating complex interventions comprising multiple, interacting components, there is a need for an evolutionary learning process to understand heterogeneities in mechanisms – what works for whom – when faced with knowledge incompleteness in the theory of change.

1. Introduction

This paper discusses the challenges of evaluating an intervention when the targeted participants have a health condition that is expected to lead to decline in health outcomes, wellbeing, and health-related quality of life. We discuss applying theory-driven evaluation to a dance intervention targeting Parkinson's disease (PD). PD is the second most common neurodegenerative disorder after Alzheimer's disease. Its cause is largely unknown, but people with PD have low levels of the brain chemical dopamine. Low dopamine levels manifest as a movement disorder with physical symptoms including rigidity, tremor, slowness of movement, impaired balance, and "freezing" – the inability to initiate movement. Other non-motor symptoms include: fatigue, depression, anxiety, cognitive impairment, masked facial expressions, sleep disturbances, and problems with swallowing, chewing, and speaking. These symptoms often lead to social isolation and lack of engagement with the world. There is no known cure for PD. However, various treatment options are available, including medications, especially synthetic dopamine taken daily. While such treatments can improve PD symptoms, they do not halt the progression of the disease.

* Corresponding author.
E-mail address: april.nakaima@utoronto.ca (A. Nakaima).

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2. Evaluation of Dancing with Parkinson's (DWP)

Dancing With Parkinson's (DWP; https://www.dancingwithparkinsons.com/) is a community-based intervention in Toronto, Canada. Since 2008 until 2020, it offered weekly, 1-hour, in-person dance classes specifically designed for individuals with PD. The Evaluation of DWP was funded by the Ontario Brain Institute's (OBI) Evaluation Support Program, which aimed to build the evaluation capacity of community organizations and to generate evidence for community-based interventions. OBI recognized that community groups such as DWP had neither the resources to commission external evaluations nor the capacity to do evaluations themselves (Nylen & Sridharan, 2020). Since the focus was on building DWP's evaluation capacity, the evaluation discussed in this paper employed strategies on a "shoestring" budget (Bamberger et al., 2004). Further details of DWP can be found in Gibson and Robichaud (2020) and Nakaima and Sridharan (2020). As part of this initiative, OBI was interested in building a community of practitioners, researchers, and evaluators to further understand how to better address issues of brain health in Ontario.

3. Towards an exploratory evaluation approach

When the population being studied is one with a *degenerative* disease such as PD, the expected trajectories are declining outcomes in health, mobility, daily functioning, and quality of life (i.e., negative slopes over time). As such, evaluating the impact of an intervention using a pre- and post- design (without a good comparison group[1]) may erroneously conclude that the intervention does not work because post-intervention outcomes have worsened – yet, this is precisely what is expected with a degenerative disease. Alternatively, one might conclude that the treatment is ineffective if there has been 'no change' pre to post – though this could actually represent success. Another challenge from an evaluation standpoint is that conventional thinking generally favors "permanent" over "transitory" changes, and "long-term" over "short-term" effects. Yet, for people living with degenerative conditions (and also people experiencing chronic pain and other chronic symptoms), transitory and short-term changes may be highly valued and welcomed relief.

Several studies have tried to understand the clinical solutions for PD (Santos et al., 2022; Avanzino & Cortelli, 2019; Sidorova & Saarma, 2020). Similarly, studies have explored whether dancing makes a difference, and if so, what kind of dance makes a difference (Dos Santos Delabary et al., 2018; Sharp & Hewitt, 2014). Despite the growth in studies, there remain significant gaps in the knowledge and practice of how best to address the challenges of PD. One major gap is understanding the mechanisms by which dance might make a difference in the daily lives of people with PD (Sridharan & Nakaima, 2020). Non-experimental studies, qualitative longitudinal studies, and better documentation of the daily lives of people with PD can help generate such knowledge. The present study is an early exploratory evaluation to better understand mechanisms and short-term outcomes.

When evaluators speak about mechanisms, or "unpacking the black box," the contents of the metaphoric box sometimes are comprised of pieces that are either unknown or "elliptical" (see Mark, 2023, this issue) or are known but difficult to define – a simple example: what is "fun"? What participants view as working for them, how they feel, and what they notice can offer clues about where to look for other evidence; but it is important not to stop there. Like other evidence and data points, clients' views, observations, and self-reports need further interrogation, confirmation, and verification.

Questions we explored included: What outcomes can be realistically hoped for and matter for people living with a degenerative condition? What are more precise framings of "improvement" or "enhancement" in a theory of change for interventions targeting degenerative conditions? While this paper does not focus on changes of a permanent or longer-term character, this does not mean that such changes are unimportant to those living with degenerative conditions. Our emphasis, however, is on short-term effects that may be transitory. Using client-driven feedback from DWP participants, this paper develops an initial but more detailed understanding of the multiple mechanisms by which engagement in dance can impact health. Our approach starts with a simplistic assumption and then moves to a more complex understanding of how participating in a program like DWP might work.

Answers to the above questions could have important implications for the recommendations of treatment plans, given that the medications taken by people with PD often have troublesome side effects, especially when higher doses are needed to manage symptoms. If dance positively impacts some of the symptoms of PD, it could be an effective companion treatment to help optimize the benefits from medication while minimizing unwanted side effects.

3.1. Our initial, simplistic assumption

Our initial theoretical stance was driven by a view of dance as exercise. There is a growing literature on the importance of the exercise/physical activity component of dance on health outcomes (Cusso et al., 2016; Mandelbaum & Lo, 2014). Likewise, numerous studies have demonstrated positive effects of exercise for individuals with PD (Goodwin et al., 2008), which have motivated the development of various exercise programs for people with PD (Heiberger et al., 2011). The following guidance from the *Canadian Guideline for Parkinson Disease* (Grimes et al., 2019) represents a dominant perspective: "*Evidence exists to support early institution of exercise at the time of diagnosis of Parkinson disease, in addition to the clear benefit now shown in those with well-established disease. It improves not only how people with Parkinson disease feel, but also their ability to perform activities of daily living.*"

Given our exploratory focus, however, we were open to exploring other mechanisms. We were aware of the possibility that dance might benefit clients through esthetic and other non-exercise related mechanisms (e.g., group interaction). The esthetic value of dance, for example, is expressed in Houston's (2015) work, *Feeling Lovely: An Examination of the Value of Beauty for People Dancing with Parkinson's*. The challenge, of course, lies in empirically demonstrating the value to health of the esthetic benefits of dance. More generally, we wanted to learn directly from the experiences of DWP clients about what made a difference to their daily routines.

Such a focus on understanding the esthetic and other benefits of dance is not just an academic concern; it has profound consequences for the guidance health organizations offer for treating people with PD. This is noted in a letter (Sack, 2020) responding to the 2019 *Canadian Guideline for Parkinson Disease*, where general exercise, surgical/-medical, and pharmacological treatments take center stage, while other movement-based therapies are sidelined:

> "*However, under the treatment section of the guideline, we would have added additional movement-based therapies for which there is solid research evidence in addition to the 2 studies of physiotherapy and occupational therapy that were referenced in the guideline. Moreover, there is a group dimension to the treatments we and others offer that can mitigate isolation and loneliness, which no doubt contribute to the risk of apathy as a barrier to patient adherence mentioned by the authors of the guideline [...].*
>
> *The panel of experts taking part in the development of the guideline is impressive. But it would be valuable to have included nonmedical practitioners, movement-based therapists and patients.*"

Exercise is predominantly conceptualized as an "activity," theorized

[1] In a better funded study, the evaluation could take measures over time from both treatment and control/comparison groups and compare the outcome trajectories. If the treatment group's decline were less steep, then the evaluation could conclude that the treatment was effective. But even such a design would not tell us about the complex mechanisms that might be involved, nor would it sufficiently address the methodological challenges of the intervention's impact outcomes over long (say 3–5 years) timelines of impacts.

in terms of its effect on muscles, connective tissue, cardiac conditioning, etc. In this paper, we also conceptualize exercise as a "mechanism" because the act of exercising brings about certain physiological changes in the body that lead to increased strength, improved balance, improved mood, etc. The assumptions about exercise as the principal mechanism at work in dancing could be rather limited, however. By seeking feedback from clients, we aimed to identify and better understand the multiple mechanisms that might be at play and learn about the limits of the predominant exercise assumptions.

The evolutionary approach we propose could be thought of from a Bayesian perspective that starts with an initial understanding of an intervention based on prior knowledge but updates this understanding as learning comes through engagement with the program stakeholders. Such a Bayesian perspective is illustrated by Dixon-Woods et al. (2011): "*The analogue we propose is that prospectively defined program theory and the program leaders' experience can be regarded as "the prior" and can be synthesized with the contribution of social sciences to produce a new ex post theory that can be used and tested in future implementations of the program.*" Instead of leaders of DWP, however, our focus was on clients. In the Discussion section, we comment on the broader implications of taking an evolutionary approach to understanding the mechanisms by which dance might benefit PD patients.

Table 1
Logic that guided our approach.

An initial model of change	1) We started with a simplistic understanding of a mechanism that might be in play. In our example we focus on *physical exercise* as an initial mechanism.
Feedback from Clients	2) We move from a simplistic understanding to a more comprehensive understanding based on client-driven feedback. Parkinson's clients who were participating in DWP provided qualitative feedback in interviews and diaries.
Seek Heterogenous Understanding of Change	3) Given the heterogeneities of symptoms experienced, stages in Parkinson's disease, and different contexts of different clients' lives, there was no reason to believe that there would be a one-size-fits-all theory of how dance might work. Our approach needed openness to understanding potential heterogeneities in what works for different clients.
Pay attention to both the short- and long-term views of change	4) Given the neurodegenerative nature of PD, the theory needs to recognize that the goal might not be long-term improvement but also short-term enhancements in health-related quality of life.
Pay attention to explicit and invisible mechanisms	5) We need to be open to the possibility of both explicit mechanisms as well as invisible mechanisms. Participants may not be aware that *touch*, for example, such as in holding hands while dancing releases oxytocin which through a neurochemical conversion can increase dopamine levels. Also oxytocin is a hormone that facilitates bonding with others, enhancing a sense of social connection.
Learning	6) Given the lack of a comprehensive theory of how dance might work, our primary focus was not to limit ourselves to testing assumptions but to learn about our initial assumptions through feedback from clients and reviewing the literature.
From simple to comprehensive	7) This paper does not develop a full comprehensive theory of dance but moves from a simplistic view of how dance can work towards a more comprehensive view that locates dance within the routine activities of participants' daily lives.

4. Methods

Table 1 describes the logic of the approach adopted in this paper. In order to study and link multiple longitudinal events to explore key linkages in the theory of change, we piloted the use of diaries (see Fig. 1). Brief entries were filled out by participants on a daily basis, with a larger set of questions completed once per week (see Table 2). The aim of the diaries was to see if even small effects could be observed on days when respondents danced versus days when they did not dance and also longitudinally over several months. There also was room for clients to provide feedback on what was working for them on any given day.

The diaries could also provide clues about the context and the routine activities of clients' daily lives – for example, some clients noted events such as going for physical therapy, meeting friends for dinner, staying at home all week. Some noted times when they struggled with fatigue, or they reflected on the timing and effects of their medication.

The daily diaries were used for up to 105 days during the summer months (mid-June to September). We anticipated that some participants would not attend classes as usual during this period (because some teachers/locations took a summer break), thus providing a stand-in for a comparison group. However, that summer all respondents attended dance classes, though not as frequently as during the regular term (September to June). Before the regular term ended, DWP teachers distributed a total of 50 diaries to participants. We anticipated a low response rate given the challenges of filling out a daily diary and so were not surprised that we received only 11 diaries. As we were trialing the use of diaries, we were not overly concerned by the small sample size. Instead, we were more interested in what patterns might emerge, if any, in terms of how clients used the diaries and in feedback from participants on improving this method of data collection – for example, clients said that filling out the diaries with pen/pencil was difficult for them and recommended typing or a daily phone call from data collectors for future iterations.

Our focus was driven partly by a gap in past investigations. Much of the research on PD has focused on what causes PD and on potential cures, with a more limited focus on enhancing daily functioning or the quality of a patient's day. Very little is known about what helps people with PD to have better days, such as combating fatigue. The main outcome measure from the diaries was in response to this simple daily question: "How are you doing today?" (See Fig. 1). Additional questions were asked once per week on Sundays (see Table 2).

This diary was complemented with a number of other evaluation elements including pre- and post-surveys, face-to-face interviews, and literature reviews. We only touch upon some of the learnings from these other methods here due to space constraints, but these other elements also added to an exploratory understanding of the mechanisms.

5. Results

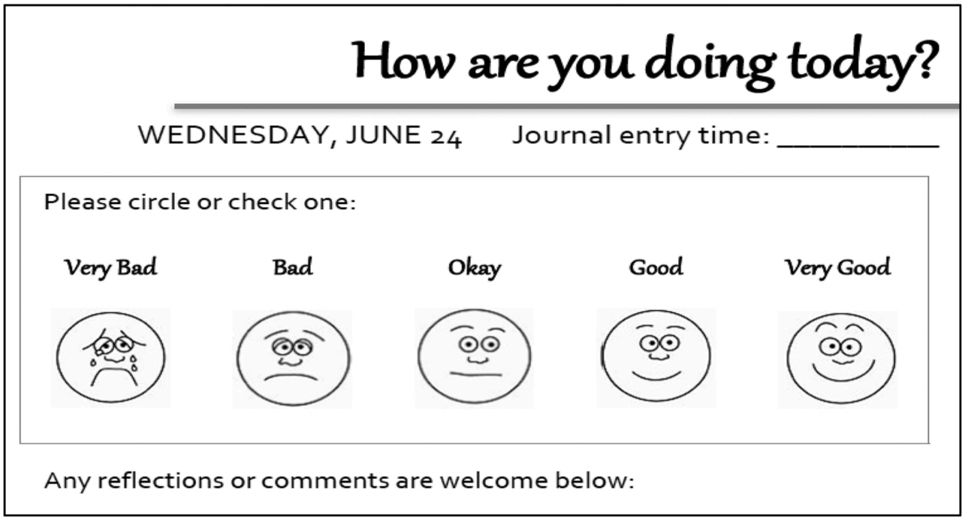
The analysis focused on three themes:

1. Exploratory learnings about mechanisms.
2. Short-term individual-level changes observed using the diary.
3. Variations between individuals – person-level characteristics that might explain variations in outcomes over time.

5.1. Exploratory learnings about mechanisms

Most of the clients support the view that the classes act as a catalyst for intentional increases in physical activity. Some comments (see Table 3) point to other mechanisms related to dance that should be considered in other studies – for example, moving one's body in a way that one is not accustomed to (see quote from respondent 3 in Table 3).

Respondent 4′s comments illustrate that the key mechanisms in a theory of change linking dance to health may very well include the


How are you doing today?

WEDNESDAY, JUNE 24 Journal entry time: ___________

Please circle or check one:

Very Bad	Bad	Okay	Good	Very Good

Any reflections or comments are welcome below:

Fig. 1. Example of daily journal page.

Table 2
Some examples of questions asked at the end of each Week.

How has your energy levels been in general over the past week?
How has your body felt in general over the past week?
How would you describe your most frequent mood in the past week?
Have you felt socially connected or supported in the past week?
How clear has your thinking seemed overall in the last week?
Looking back on the week, did you happen to do any dancing or exercise?
Any reflections on the week are welcome below:

interplay between the inspirational mechanisms of hope, freedom, joy, vitality, and social connection. Respondent 4′s view was shared by most respondents. This insight is especially important in moving beyond a view of dance as primarily being driven by a mechanism of exercise: *"But it works. There are some things that are possible with dancing that are not possible with exercise. […] That's one thing I discovered with Parkinson's, it's not just physical, it's emotional. It's emotional highs as well, which is good."*

Another client we interviewed (respondent 5) who had been attending DWP for 8 years said that the music and social connections he gets through DWP makes a difference. But in response to our follow-up question (*And have you found any benefits to your health as a result of this*

Table 3
Examples of Quotes from Respondents.

Respondent number	Quote
1	"*It's something I can't do without. And if I miss, there's all hell to pay in my body – it just doesn't feel as good. … It's necessary for my existence, to have this.*" This client came on a wheelchair scooter two miles in the rain to attend the class that day and had been attending for 8 years.
2	*"Oh, it helps, it really helps me. I was able to lower one of my medications. … I find that the movement – and I do some of the steps, some of the little mini parts at home. … I swing my arms and I think of [dance teacher]. I am probably a bit more agile than a year ago. I am definitely… I get out of bed better. And by doing these kinds of exercise and movements and… it encourages me."*
3	*"I come back for a number of reasons. First of all, it's great exercise and it's exercise in a very informal setting and I like the fact that it's exercise to music. And I find that moving my body in ways that I'm not accustomed, although they're quite natural because they're dancing, helps me with the illness, or helps me cope with the illness. It makes me feel better. And I think it has some physical effects on the Parkinson's illness itself, too. I have said this before, one of the most surprising things, and the most significant things is the humanity I find here […] to see how people cope with this terrible malady, and enjoy themselves to music, I think suggests a kind of human connectedness that is very real and very important. […] I look forward to it and I try to do a little dancing at home too.*"
4	"*It's a happy time. So you come for happiness. And we've all got diseases, various forms of the disease, Parkinson's, but it's a fun time to be with other people. […] But it also … I shouldn't be able to do the things I do. But it works. There are some things that are possible with dancing that are not possible with exercise. […] That's one thing I discovered with Parkinson's, it's not just physical, it's emotional. It's emotional highs as well, which is good. You can get a high on music. And that conquers Parkinson's as well, tackle it on all fronts type of thing. It's strange because I was trying to do, learn some line dancing, international dancing. And I'd fall, I'd tend to fall, because I'm doing these 1, 2, 3, 4, 5 – not to music. But as soon as they put music on, I can anticipate, somehow anticipate the next note. And it keeps me from falling. And I can do this extensively. […] And you just have enjoyment like this. And then you let the music work with you. And you, you almost, it's a high! And the endorphins are running around and creating all sorts of activities. And it's great! […] And I can't wait to, for the daytime so I can go and do all this again. So it's an encouragement thing. […] It's a strange thing, but there's some connection. There's some connection that involves: Exercise, Music, and Parkinson's. And hopefully you guys and other people working in the business will discover that.*"
Respondent number	Quote
5	*"No, too difficult to say. You know you have to take all the medication and do everything else you are told. Some things help a little bit, some others don't, I suppose. But we all are optimistic that everything helps. I am going to 2 classes; I am going to Wednesdays and Saturdays. And on a day like this, today, it was not a good day because in the weather to get out, but here I am. Well, I like to dance, before I had Parkinson's, I like to dance too."*
6	*It has occurred to me that changes, like "okay" to "good" are more the result of medication variations (frequency* etc.*) than of DWP classes. Yesterday was the last DWP class of the season. [Dance teacher] asked me at the start of class what I would like for the finale of the class, and I said a cha-cha lesson. She complied. At my departure, she gave me a delicious hug. Hugs are so important. I could write an essay on them." (respondent 6, July 3)* *"I felt good today except around 1–3 pm during which I felt lousy. It's due to poor management of my medications. Again, I want to stress that I believe this is a much bigger factor in my state of well-being than whether I've had a DWP session. My daughter (age 21 and a psych major at [a high-ranked university]) just made a remark that rings very true to me. She said that DWP sessions don't have a noticeable effect on a day-to-day basis, but she strongly believes that I'm happier now than I was before I began DWP sessions. For the reasons above, I didn't continue the journal after here." (respondent 6, July 6)*

dance?), he said: *"No, too difficult to say. You know you have to take all the medication and do everything else you are told. Some things help a little bit, some others don't, I suppose."* Like respondent 5, a few other clients provided comments in their daily diaries that suggest they were unaware of the direct physical health benefits associated with DWP, despite excellent records of their regular weekly attendance. These respondents commented that better or worse days probably had more to do with their medication management/timing than with dancing. For example, one very active client (he exercises for more than one hour 6–7 days per week) noted in his last diary entry that he found the diary *"quite useful. What I observed in myself over the past 15 weeks is that I felt Bad or Okay only when I was late with my medications and/or did not get enough sleep. I felt Good or Very Good when I had done some exercise, including but not limited to DWP."* He goes on to discuss his medication schedule and great support from his wife, family, instructors (personal fitness, Pilates, Qigong) in for the past 12 years. He concludes with this: "*Finally, I must say that DWP on Wednesdays is often the highlight of my week. In large part this is because of the wonderful, caring teachers and volunteers, over the past seven years [he names 7 of them] …Thanks to all of them.*"

Another respondent (see Table 3, respondent 6) concluded in his first week of diary entries that DWP did not have a *"noticeable effect on a day-to-day basis."* Based on this conclusion, he decided to discontinue the daily diary entries, thus eliminating the opportunity to see, as we saw with other respondents, whether there was a relationship between higher-rated days and DWP class days. Yet, as the two journal entries in Table 3 hint, there are multiple mechanisms at play – e.g., hugs (touch) and joy.

This set of responses provides some clues on how dance might matter for DWP participants on a daily basis and also the utility of a diary in surfacing potential mechanisms and valued outcomes. We view this as a process of building some initial knowledge of the mechanisms by which dance may work. As we highlight in the Discussion section, an evolutionary approach can build knowledge of the short-term linkages related to dancing with health and wellbeing outcomes, as well as learn from practitioner-based innovations such as Dancing with Parkinson's. A diary such as we trialed provides one input to understanding daily client experiences over time.

5.2. Variations in individual-level outcomes over time

We also utilized the diaries to learn how experiences varied over time in the short-term. Stated differently, focusing on the diaries, we wanted to find out if participants noted feeling better on days that they danced. And if so, did the good feeling last for a few hours, one day, or longer? How did DWP days compare with the rating on other days when respondents went to physical therapy, the gym, swimming, music concerts, or were inactive at home, etc.? Fig. 2 provides an example of one individual's diary comments along with the corresponding rating in response to the question: *"How are you doing today?"* Over the course of approximately 105 days, this person responded as follows: "bad" 4% of the time, "okay" 91% of the time, and "good" 5% of the time. Notably, the respondent's highest ranked days aligned with DWP class days, with the exception of one DWP day where the rating was just "okay" however, in this instance, the diary entry was completed prior to attending DWP that day: *"DWP begins today – looking forward."* Curiously, with this same individual, some of the days that were rated as just "okay" were days when the respondent attended concerts and birthdays.

Fig. 3 does the same for another individual. Both Figs. 2 and 3 demonstrate how we attempted to link the analysis of the outcome data (i.e., how the client is doing on a given day) with other life events.

The responses in Fig. 3 clearly indicate that the respondent pays attention to their activity level. It is notable that this client attributed the elevated mood that carried over to the next afternoon to his DWP class the previous day, and he also claims that the effects from dancing and the stimulating music helped him get through a long, tedious medical procedure.

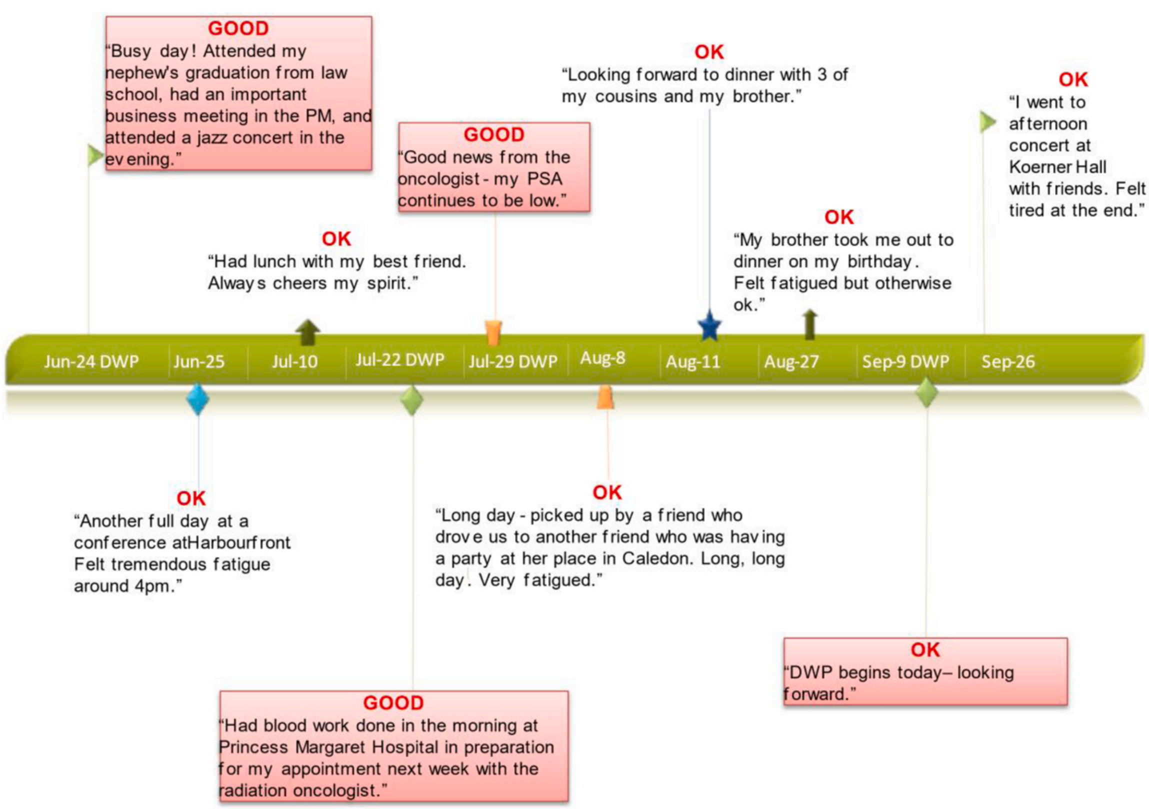


Fig. 2. Individual-level variations for Client X.

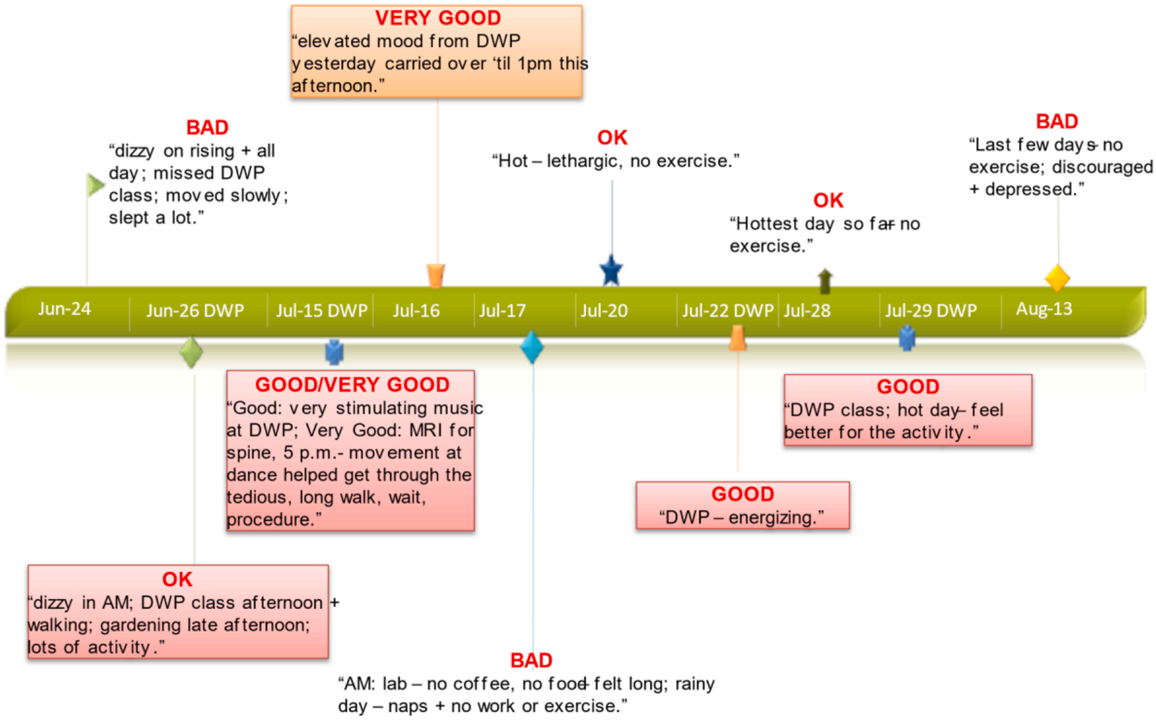


Fig. 3. Individual-level variations for another Client Y.

5.3. *Variations between individuals–person-level characteristics that might explain variations in outcomes over time*

The small sample size limited our ability to conduct analyses examining the variations among individuals in the dynamic linkages. In other words, did dance lead to short-term improvements only for some individuals? If so, what are the characteristics of such individuals? The realist evaluation mantra of 'what works, for whom, in what circumstances?' (Pawson & Tilley, 1997) could be re-phrased more dynamically by: 'what works, for whom, in what circumstances and *when*?' This approach could help us learn how the dynamic effect of dance differs across individuals. We discuss the relevance of this question for an evolutionary approach to knowledge building in the Discussion. The strength of such an analysis is that individual and time-varying factors can be used to account for changes in daily variations and outcomes. Given our sample size of 11, we are unable to conduct an analysis but can outline potential avenues by which a larger sample size could unpack the assumptions linking engagement in dancing to improved wellbeing. First, following the analysis scheme of calendar data done by Axinn et al. (1997, 2019), multilevel models can be used to study both between-individual differences and time-varying factors to explain the variations in outcomes. This first approach can be further complemented by methods of person-centered developmental trajectories (Hofmans et al., 2018) which examine differences in the role of various factors across heterogenous groups/segments of individuals. Additionally, qualitative longitudinal research (Calman et al., 2013) methods can help in understanding heterogeneities in mechanisms within the journeys of different clients.

6. Discussion

This evaluation highlighted the potential of diaries to better understand the short-term experiences of clients using their daily routines as a means of learning about potential mechanisms and what matters to clients. Larger and more rigorous studies are needed to build evidence from client experiences on what works, for whom, and when.

6.1. *The utility of diaries*

Diary data can deepen our understanding of the *'architecture'* of clients' lived realities, such as how a client's day plays out when they feel energized or how well they manage their illness. The present analysis was limited by a small sample size. However, based on our initial findings, the collection of diary information from larger samples of people with PD could offer rich insights into the complex interplay between an intervention like DWP, prescription medications, other treatments, and life events that can impact the energy and wellbeing of a client on any given day. Such diaries could provide a "window" into clients' daily routines and enable more nuanced insights into the heterogenous mechanisms by which greater wellbeing can be achieved.

Future applications of diaries can also consider advances in sampling techniques such as time-sampling studies (Mann et al., 1991) and also collecting the data through phone or other electronic devices.

6.2. *Incompleteness in knowledge: dance as part of a larger 'causal bundle'*

Knowledge of the causal structure connecting engagement in dance to improved health is largely incomplete. While the literature review conducted for this study unearthed some important research on the impacts of dancing on health, further clarity is needed regarding an overall causal theory of the mechanisms by which dancing can impact health. The programmatic and methodological challenge is not to simply arrive at an estimate of the direct effect of dancing on health; the challenge is far bigger.

The larger "causal bundle" (Mayne, 2012—note Mayne prefers the term *causal package*; Cartwright & Hardie, 2012) that connects engagement in dancing to improved health requires further explication. In the absence of precise theory, we adopted an empirically driven strategy to learn about other potential 'conjunctive' factors that might enhance the impact of dance on clients' health, daily functioning, and overall wellbeing. Based on the diary data, we learned, for example, that dancing can provide a boost in energy that enabled some clients to engage in other routine activities without fatigue. While such routine activities might not sound dramatic, from the client's perspective, this could mean the difference between having a good day versus feeling fatigued or depressed.

Our allegiance should not be to predetermined outcomes; we need to create room for clients living with chronic, neurodegenerative conditions like Parkinson's to articulate their own priorities – what matters to them, what gives them energy, and how an intervention like DWP can be helpful. We believe such a focus can also help generate insights on the multiple mechanisms by which dance can improve health. If we consider an outcome such as improved quality of life, dance can be viewed as an INUS cause (Mackie, 1974) of improved quality of life. By this we mean, dance by itself is an *insufficient* but *non-redundant* part of an *unnecessary* but *sufficient* condition for improved quality of life. The interplay between dance and other factors such as life events, other interventions including medications, needs to be more fully understood. Understanding both the other conditions needed for dance to work and the heterogeneities of other factors at play should be the focus of future research and evaluation of DWP.

6.3. Towards an emergent understanding of the relationship between dance and health

In our view, it is important that any exploration of causal pathways connecting engagement in dance to improved health respect the heterogenous journeys of clients with PD. In this light, it becomes especially critical to learn more about the short-term daily experiences of clients from the perspectives of clients themselves. Our initial exploration was based on the predominant view that the impact of dancing on health and wellbeing is primarily due to its physical exercise component. However, feedback from the diaries and qualitative interviews suggest that a wider set of mechanisms are involved. The theoretical challenge for the field is not only in cataloging the different mechanisms by which a complex program like DWP can make a difference in the lives of clients, but also in understanding the interplay between the different mechanisms in the progression and management of PD for different types of clients.

6.4. Realist evaluation as an approach to building knowledge

Along with exercise, other types of activities and mechanisms might matter for different individuals. Realist evaluators (Pawson & Tilley, 1997; Pawson, 2013) argue that mechanisms activate only under some contexts. One area for future exploration is to better understand the conditions under which a dance class can be helpful. Understanding the specific context-mechanism-outcome configuration is one important focus of realist evaluation. Rather than move to a test of a simple assumption model, the exploratory approach we advocate moves towards broadening the range of assumptions under consideration. In our view, the diaries were useful in better understanding the multiple mechanisms at play.

6.5. The challenge of incompleteness in knowledge: looking to evolution for inspiration

The state of knowledge linking dance to health is incomplete. In their seminal paper, Lieberson and Lynn (2002) argue that some of the most complex knowledge building activities in the social sciences can look to evolution for inspiration to address incompleteness in knowledge:

> "The evolutionists do not confuse fatal errors, on the one hand, with problems stemming from incompleteness, information that is still insufficient or not yet determined, or even unresolved. The latter cases are worrisome and certainly not to be glossed over. Yet, it does not necessarily mean that the theory is to be abandoned or that Darwin was wrong. In evolution, incomplete is not the same as erroneous" (p. 4).

Their argument is that we need to depend far more on observational and longitudinal studies to better understand complex laws that govern social and other health phenomena. Lieberson and Lynn (2002) also argue for a pluralistic view of causation – we need to go beyond narrow views of what constitutes a cause in understanding social and individual change. They suggest looking to evolution to build a better understanding of multiple causes:

> "[…] as Mayr (1961, p. 1503) observes for his own discipline, many heated debates about the cause of a phenomenon 'could have been avoided if the two opponents had realized that one of them was concerned with proximate and the other with ultimate causes.' Evolution does not view all causes as potentially equal, but rather operates with a causal hierarchy so that broader conditions are necessary for narrower conditions to operate. As a consequence, it is important to distinguish tightly between basic and nonbasic causes. It would be terribly helpful for us too" (p. 12).

As we seek to understand a scheme by which dance, medicine, and other treatments can be combined as part of multiple 'causal packages,' we think it may be useful to focus on the potential utility of dance to improve the short term (e.g., quality of the day) and medicine's and other clinical treatments' ability to impact the longer term. A focus on multiple causes and a pluralistic understanding of causation (Davidoff, 2009) can help with such integration.

6.6. Celebrating John Mayne: towards an evolutionary approach

In much of our work, we have learned from John Mayne; John was a key author in a prior forum in the journal *Evaluation and Program Planning* in 2017 on lessons learned on building evaluation capacity for brain health in Ontario. John was not just a good evaluation methodologist who cared about theories of change; he also understood the vital importance of building capacities within community groups. His paper for the prior volume was titled, "Building Evaluative Culture in Community Services: Caring for Evidence."

Our paper suggests that we need to care about evidence of very different kinds, including what it takes for clients to have a good day. As we build a community that helps develop knowledge about what works for people with PD and learns from successful programs in clinical and community settings, we move beyond the view of knowledge developers as only academics or researchers to greater attention and regard for the innovations happening within communities. While there will be a need for thinking about epistemologies of what works through broader lenses, this focus on research and methodological innovation needs to be complemented with a recognition that there's already much excellence happening in community settings and the routine lives of patients. A respectful and opportunistic interplay between formal evidence, theory, and experience is needed as we build a learning community.

Our prior work has argued for an evolutionary process (Sridharan et al., 2016, p. 96) in understanding heterogeneities in mechanisms (what works for whom) when faced with knowledge incompleteness in the theory of change:

> "The first phase would focus on learning about both the context and mechanisms (and the need for heterogeneous mechanisms) and the second phase would be a testing phase in which the revised and refined program that seeks to address the heterogeneous needs of individuals is tested. A subsequent phase can test the revised program theory in a more elaborate way. We note that such an iterative, evolutionary, adaptive stance to evaluation design has been the focus of a literature on sequential and adaptive designs [.]; however, this literature has not been explicitly connected to the literature on theory of change and does not focus on learning about heterogeneous mechanisms."

There is a real opportunity to build knowledge connecting dance to health by focusing on the following areas:

1. Building community – A key strength of the Ontario Brain Institute initiative was the opportunity it provided to build community by involving clinicians, patients, policymakers, community-based programs like dance, researchers, and evaluators. Future evaluation and research studies would be beneficial if embedded in such community-building activities.
2. Knowledge translation and linking dance with clinical interventions – How can dance be integrated into medical/clinical solutions by neurologists?
3. Short-term changes – Given that PD is a neurodegenerative disease with no known cures, what helps create a good day for someone with PD? Non-experimental studies and qualitative longitudinal studies of people with PD can help generate this knowledge.
4. Heterogeneities in what works and in what "it works" means – It is entirely possible that different causal bundles might work, and perhaps in different ways for different individuals. For clients, what is the causal bundle that corresponds to a good day?

Taking an evolutionary approach suggests evaluation purposes beyond summative in helping build knowledge of heterogenous causal bundles. This might include planning for a portfolio of evaluations with various purposes and designs to answer sets of questions and to progressively incorporate and verify knowledge as it is generated. We need to learn from assumptions and better understand the contexts under which the assumptions might hold.

7. Conclusions

A critical learning from this evaluation was the need to rethink what success means when faced with a neurodegenerative disease. The focus of an intervention such as DWP is not to cure PD. Instead, the focus is to ensure that clients feel better daily and feel energized to function through the week and perhaps slow the progression of PD. Physical exercise may be one mechanism by which clients experience a boost in energy; yet, other mechanisms, such as touch, feeling stimulated by the music, and esthetics including "feeling lovely" (Houston, 2015) may have important linkages to health and be fundamental to clients' well-being. Along with other chronic conditions such as diabetes, pain, and depression, it is not just about waiting to get rid of the problem, but also importantly about daily management of the complications and living better with the condition.

Declarations of interest

None.

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April Nakaima is a Research Officer at The Evaluation Centre for Complex Health Interventions at University of Toronto in Canada. She has co-led evaluations of poverty reduction initiatives, brain health community initiatives, and dance activity health promotion. April has co-edited a previous volume in *Evaluation and Program Planning* on

building evaluation capacity with community organizations, in *New Directions for Evaluation* on addressing health inequities in China, India and Chile, and on evaluator competencies (currently in press) in the *Journal for MultiDisciplinary Evaluation*. April is a graduate of Kamehameha Schools in Hawaii and the University of California, Santa Cruz and Irvine.

Sanjeev Sridharan is Professor of Health Policy Evaluation at the University of Hawaii at Manoa. Previously he was the Country Lead, Learning Systems and Systems Evaluation at the India Country Office of the Bill and Melinda Gates Foundation. Sanjeev was the Director of the Evaluation Centre for Complex Health Interventions and Associate Professor at the Department of Health Policy, Management and Evaluation at the University of Toronto. He is a former associate editor of the *American Journal of Evaluation* and has been on boards of the *Canadian Journal of Program Evaluation, New Directions for Evaluation* and *Evaluation and Program Planning.*

Rachael Gibson, PhD, is a Senior Evaluator at the Evaluation Centre for Complex Health Interventions at the Dalla Lana School of Public Health at the University of Toronto. For over a decade, Rachael has been working on health promotion for marginalized or disadvantaged individuals, including seniors living with Parkinson's Disease, individuals experiencing homelessness, mental health issues, and addictions in a drop-in setting, as well as individuals and communities coping with gender-based violence. Rachael also has experience in knowledge exchange/transfer regarding how to improve and sustain programs. Rachael has an academic background in political economy and economic geography.